Jane's Candy Bowl Executive Summary

Intro/ Hook

A candy bowl sits on most counselors' desks, but many students barely notice. What if we changed that? With the claw machine candy bowl from the Joseph Garbini Design Firm, we have implemented both interactiveness and engagement into the standard candy bowl to make grabbing candies more fun and be a conversation starter for students visiting.

Problem/Context

This project aims to bring in more students to Jane's (an academic counselor for master's students at UW department of Mechanical Engineering) office, so we plan to achieve these objectives with the mini claw machine design. While standard candy bowls are forgettable, this candy bowl requires the user to work for their candy in a unique and challenging way that will get them excited when they are finally able to grab it.

Objectives

The selected design aims to prioritize an interactive experience with a user's interaction time of 2-3 minutes and a difficulty level estimated for a 80% success rate on their third attempt. To ensure usability in office or career fair environments, the system is required to fit in an approximately 12x12x12 inch space. The machine's reliability is marked to not fail until at least 200 cycles and not to break while being carried or transported.

Solution/Overview

Our reimagined candy bowl is a claw-machine system that enhances user engagement while remaining simple to operate and maintain. The system features a 2-axis sliding gantry for precise positioning and a hydraulic syringe actuator that delivers the necessary grip force to retrieve small treats like Starbursts. The entire assembly is housed within a transparent acrylic enclosure, ensuring safety and durability while maintaining full visibility of the mechanism in motion.



Figure 1: Sliding Gantry

To optimize the scissor lift power and claw ability, the team decided to use water for the hydraulic mechanism. The structure is built from lightweight makerbeams that are not only easy to assemble, but also convenient to be attached securely with acrylic panels. Furthermore, the scissor lift mechanism employs a double-nut locking technique to ensure smooth pivotal motion without the risk of fasteners loosening over time.

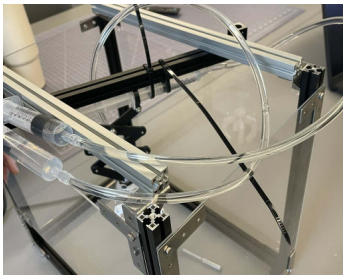


Figure 2a: Hydraulic Power

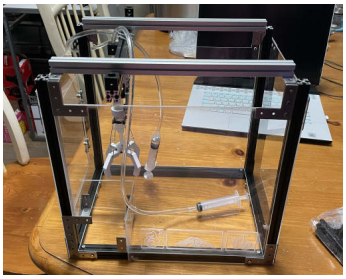


Figure 2b: Makerbeam Structure



Figure 2c: Double-nut Technique

Results:

The makerbeam structure ensures the device is sturdy yet lightweight, approximately 3lbs. Additionally, with 300mm (12-in) beam length, the machine maintains a compact base area with the dimensions satisfying the requirement to fit on Jane's office desk. And most importantly, live testing has been run on 15 users, where 11 of them successfully retrieved the first candy in their third attempt with an average interaction time of 2 minutes. Of these users, 10 of them commented on the design fun to use, which confirmed the design's engagement goals (fun, interactive, and challenging).

Impact & Recommendations

The goal of this design was to increase the engagement of Jane's students as they visited for counseling and during career fairs. With the design being fun and attracting, it brings curiosity around the candy bowl and incentivises students to interact with it. For the future, it would be good to continue testing to ensure that the claw hydraulics does not leak and works consistently every time and gather feedback from students to further improve on the design.